# 3.1 Law of Sines

Name _____ Period ____ Date _____

Pre-Calculus 3.1 (30-minute plan) · Student sheet

**I can…**

- use the Law of Sines to solve oblique triangles (AAS or ASA).
- use the Law of Sines to solve oblique triangles (SSA), including the ambiguous case.
- find the area of an oblique triangle and solve real-life problems.

**Vocabulary** · fill in the missing word

**Oblique triangle** — a triangle with no _____ angle

**Law of Sines** — a/sin A = _____ = c/sin C

**Ambiguous case (SSA)** — two sides and an angle _____ one of them: there may be no triangle, one, or two

**Area of an oblique triangle** — Area = _____ = ½ab sin C = ½ac sin B

## Part 1 · Notes and examples (fill in each note, then work the example)

**Law of Sines:** _____ = _____ = _____

**When to use it:** Use the Law of Sines for _____, _____, and _____.

### Example 1

Solve the triangle: C = 102.4°, B = 28.7°, b = 27.4 feet. Round to two decimal places.

**The ambiguous case (SSA):** In the SSA case, if sin B > 1 there is _____. If sin B < 1, check whether _____ also works.

### Example 2

Find two triangles with A = 20.5°, a = 12, and b = 31. Round to two decimal places.

**Area** of an oblique triangle = _____

### Example 3

Find the area of a triangular lot with sides of 52 meters and 90 meters and an included angle of 102°.

## Part 2 · Partner practice

### Partner practice

How many triangles have A = 30°, a = 6, and b = 10? Find B for each.